

PRAKTIKUM SISTEM OPERASI

MODUL 10

SHELL



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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall.

Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h chname diubah menjadi:

```
extern status ascdade(uint32, char *);
/* in file bufinit.c */
extern status bufinit(void);
/* in file chprio.c */
extern syscall chprio(pid32, pri16);
/* in file chname.c */
extern pri16 chname(pid32, char *, uint32);
/* in file clkupdate.S */
extern uint32 clkcount(void);
```

- b. Pada chname.c fungsi diubah dari:

```
#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
syscall chname(
    pid32      pid,
    char *      newname
)
```

menjadi

```
#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
pri16 chname(
    pid32      pid,
    char *      newname,
    uint32      namelen
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil

```

pri16 chname(
    pid32      pid,
    char       *newname,
    uint32     namelen
)
{
    intmask mask;
    struct proctent *prptr;
    uint32 i;

    mask = disable();

    /* Validasi PID */
    if (isbadpid(pid)) {
        restore(mask);
        return SYSERR;
    }

    prptr = &proctab[pid];

    /* Nama baru */
    for (i = 0; i < namelen && i < PNMLEN-1; i++) {
        prptr->prname[i] = newname[i];
    }

    /* Null terminator */
    prptr->prname[i] = '\0';

    restore(mask);
    return OK;
}

```

2. Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell!

a. Edit shell.c

```

#include <xinu.h>
#include <stdio.h>
#include "shprototypes.h"

/*****
 * Table of Xinu shell commands and the function associated with each
 *****/
const struct cmdent cmdtab[] = {
    {"argecho", xsh_argecho},
    {"arp", xsh_arp},
    {"cat", xsh_cat},
    {"clear", xsh_clear},
    {"date", xsh_date},
    {"devdump", xsh_devdump},
    {"echo", xsh_echo},
    {"help", xsh_help},
    {"ls", xsh_ls},
    {"kill", xsh_kill},
    {"memdump", xsh_memdump},
    {"memstat", xsh_memstat},
    {"ns", xsh_ns},
    {"netinfo", xsh_netinfo},
    {"ping", xsh_ping},
    {"ps", xsh_ps},
    {"sleep", xsh_sleep},
    {"tee", xsh_tee},
    {"udp", xsh_udpdump},
    {"udpecho", xsh_udpecho},
    {"udpserver", xsh_udpserver},
    {"uptime", xsh_uptime},
    {"namecmd", xsh_namecmd},
    {"mycmd", xsh_mycmd},
    {"?", xsh_help}
}

```

b. Edit shprototypes.h

```

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);

```

c. Buat file baru:

3. Test hasilnya:

- Masuk ke terminal xinu
- Jalankan perintah ps

```
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

- Jalankan perintah namecmd

```
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Jalankan perintah ps

```
Welcome to Xinu!

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192 0 0
6 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $ namecmd
My new command
xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr  Stack Size Msg  Content
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFECC 8192 1 3
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 hello       wait   500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell       recv   50   4 0x005F7FFC 0x005F79AC 8192 0 7
8 ps         curr   20   5 0x005F5FFC 0x005F5FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:0 | ttyS0
```

- Lihat nama proses telah berubah

Nama proses berubah dari *netin* menjadi *hello* karena syscall `chname()` berhasil dijalankan melalui command `namecmd`.